THE CHOW RANK OF THE SIX-BY-SIX PERMANENT

RONGYU XU

ABSTRACT. Let perm_6 be the permanent of a generic 6 by 6 matrix. Over every field of characteristic zero, we prove that its ordinary Chow rank is 32. The finite computations are stated as explicit lemmas with exhaustive domains and exact outputs. The lower bound combines a symmetric image-span inequality with a half-defect estimate for arbitrary quotient derivative symbols of a sextic Chow term. An exact verification reconstructs all finite tables. A second implementation independently checks every labelled bipartite graph with at most six edges and all 45,696 coordinate squarefree-symbol cases. No floating-point or randomized step is used. A separate standard-library implementation independently expands and differentiates all five dependent-factor normal forms. No border-rank or general-n equality is claimed.

1. INTRODUCTION

The Chow variety in degree six parametrizes products of six linear forms. Its secant geometry leads to two distinct invariants: ordinary Chow rank, which requires an actual finite sum of products, and border Chow rank, which allows limits of such sums. This article concerns only the ordinary invariant. Product rank is synonymous with ordinary Chow rank in the relevant literature; see [3].

Glynn's identity gives a 32-term product decomposition of perm_6. The content of Theorem 2.2 is the matching lower bound. The proof uses the middle-to-quadratic derivative tower, a small-intersection theorem for permanent quadrics, and a local half-defect estimate for every sextic Chow term. A symmetric image-span inequality and a filtration of the factor spans give two estimates with the same middle-rank defect term. Section 7 defines every quantity in those estimates; cancellation of their common defect forces at least 32 summands.

The local half-defect estimate contains the finite part of the argument. It uses exact rational differentiation of the five dependent-factor normal forms, an exact enumeration of 45,696 coordinate symbol cases, and a coordinate verification of the small permanent-quadratic intersections. The verification reconstructs these objects from their definitions and compares them with the

Date: October 16, 2026.

2020 *Mathematics Subject Classification.* Primary 15A69; Secondary 14N05, 68V05.

Key words and phrases. permanent, Chow rank, product rank, catalecticant, computer-assisted proof.

exact payload. All geometric reductions and global inequalities are written proofs.

2. Statement and permanent derivatives

Notation and conventions. Let $\mathbb{K}$ be an algebraically closed field of characteristic zero. For a positive integer n, write $[n] = \{1, 2, \dots, n\}$ and let S_n be the group of bijections of $[n]$. If W is a finite-dimensional $\mathbb{K}$-vector space, then W^*, $\operatorname{Sym}^j W$, and $\operatorname{Gr}(a, W)$ denote its dual, its degree-j symmetric power, and the Grassmannian of a-dimensional subspaces. For a finite set Ω, $\binom{\Omega}{a}$ denotes the set of its a-element subsets. For a subspace $H \subset W$, its annihilator is $H^\perp := \{\lambda \in W^* : \lambda(h) = 0 \text{ for every } h \in H\}$. We use span, im, ker, rank, and dim with their usual linear-algebraic meanings.

Let

$$V = \operatorname{span}_{\mathbb{K}}\{x_{rc} : 1 \le r, c \le 6\}, \qquad \dim V = 36.$$

Write $X = (x_{rc})_{r,c\in[6]}$ and define

$$\operatorname{perm}_6 = \sum_{\sigma\in S_6} \prod_{r=1}^{6} x_{r,\sigma(r)} \in \operatorname{Sym}^6 V.$$

For $f \in \operatorname{Sym}^m V$, $0 \le a \le m$, and $\lambda_1, \dots, \lambda_a \in V^*$, let $\partial_{\lambda_1} \cdots \partial_{\lambda_a} f$ denote iterated directional differentiation. With $j = m - a$, define

$$\begin{aligned} C_{a,j}(f) &: \operatorname{Sym}^a V^* \longrightarrow \operatorname{Sym}^j V, \\ C_{a,j}(f)(\lambda_1 \cdots \lambda_a) &:= \partial_{\lambda_1} \cdots \partial_{\lambda_a} f, \\ \mathcal{D}_j(f) &:= \operatorname{im} C_{a,j}(f). \end{aligned}$$

For $G \subset \operatorname{Sym}^j V$, put $\partial G := \operatorname{span}\{\partial_\lambda g : \lambda \in V^*,\ g \in G\}$, and write $\partial g := \partial \operatorname{span}\{g\}$. For $g \in \operatorname{Sym}^j V$, let $\operatorname{Ess}(g)$ be the smallest subspace $L \subset V$ for which $g \in \operatorname{Sym}^j L$. Its dimension is the number of essential variables of g. For $q \in \operatorname{Sym}^2 V$, let $M(q)$ be the symmetric matrix of its associated bilinear form in the ordered basis (x_{rc}), and define the quadratic rank of q to be $\operatorname{rank} M(q)$. Finally, put $E_j := \mathcal{D}_j(\operatorname{perm}_6)$.

For $I, J \subset [6]$ with $|I| = |J|$, let $X_{I,J}$ denote the corresponding submatrix of X, and let $\operatorname{perm}(X_{I,J})$ denote its permanent, defined by the same permutation formula as above.

Definition 2.1. A sextic Chow term is a nonzero product $T = \ell_1 \cdots \ell_6$. The ordinary Chow rank $\mathrm{R}_{\mathrm{Chow},\mathbb{K}}(f)$ of $f \in \operatorname{Sym}^6 V$ is the least positive integer N for which $f = T_1 + \cdots + T_N$ with sextic Chow terms $T_1, \dots, T_N$; we set $\mathrm{R}_{\mathrm{Chow},\mathbb{K}}(0) = 0$.

Theorem 2.2. *Over $\mathbb{K}$,*

$$\boxed{\mathrm{R}_{\mathrm{Chow},\mathbb{K}}(\operatorname{perm}_6) = 32.}$$

The same equality holds over every characteristic-zero field.

We first record the permanent derivative facts used below.

Lemma 2.3 (Permanent derivative tower)**.** *One has*

$$\dim E_3 = 400, \qquad \dim E_2 = 225, \qquad \partial E_2 = V,$$

and

$$E_2^{(1)} := \{g \in \operatorname{Sym}^3 V : \partial g \subset E_2\} = E_3. \tag{2.1}$$

Moreover, every nonzero element of E_2 has quadratic matrix rank at least four, and every nonzero element of E_3 has at least nine essential variables.

Proof. The 3×3 and 2×2 subpermanents are bases of E_3 and E_2: distinct row-column blocks have disjoint matching supports. Their numbers are $\binom{6}{3}^2 = 400$ and $\binom{6}{2}^2 = 225$. Differentiating a rectangle permanent at one corner gives the opposite variable, proving $\partial E_2 = V$.

The inclusion $E_3 \subset E_2^{(1)}$ is immediate. Conversely, suppose all first derivatives of a cubic g lie in E_2. Row and column multiaffinity forces every monomial of g to be a three-edge matching. Inside a fixed 3×3 block, the rectangle relations equate coefficients of matching monomials related by a transposition. Transpositions connect all six matchings, so each block is a scalar subpermanent. This proves (2.1).

The row-column diagonal torus has distinct characters on each subpermanent basis. A generic one-parameter subgroup degenerates any nonzero element to a single basis element, while matrix and first-catalectic ranks cannot increase under specialization. A rectangle permanent has quadratic rank four. The nine first derivatives of a 3×3 permanent are independent. The two rank floors follow. □

3. Small permanent-quadratic intersections

Lemma 3.1. *If $L \subset V$ has dimension at most six, then*

$$\dim(E_2 \cap \operatorname{Sym}^2 L) \leq 3.$$

If $\dim L \leq 5$, the upper bound improves to one.

Proof. For fixed nonnegative integers a, q, the locus in $\operatorname{Gr}(a, V)$ on which $\dim(E_2 \cap \operatorname{Sym}^2 L) \geq q$ is closed, projective, and stable under the row-column torus. Choose a one-parameter subgroup with distinct weights on the 36 variables. The projective limit of any point of this closed locus is fixed by the subgroup. Its variable eigenspaces are one-dimensional, so the limit is a coordinate a-plane, equivalently a bipartite graph with a edges. This elementary torus specialization is the only fixed-point argument required here; compare the general fixed-point framework in [1].

At a coordinate point, the intersection has one independent rectangle permanent for each four-cycle. Two distinct four-cycles have a union of at least six edges. If the union has exactly six edges, it is $K_{2,3}$ or $K_{3,2}$, which has exactly three four-cycles. Thus five edges support at most one four-cycle and six edges support at most three. □

4. Local linear algebra

If $L \subset V$ and $(y_1, \dots, y_\ell)$ is a basis with dual basis $(y_1^*, \dots, y_\ell^*)$, define the polarization map

$$\delta_L : \operatorname{Sym}^3 L \longrightarrow L \otimes \operatorname{Sym}^2 L, \qquad \delta_L(g) = \sum_{j=1}^{\ell} y_j \otimes \partial_{y_j^*} g.$$

This definition is basis-independent and δ_L is injective. We write δ when L is clear.

Lemma 4.1 (Kernel-preimage bound)**.** *Let D be a vector space, let $P : L \twoheadrightarrow D$ be a surjective linear map of rank $d = \dim D$, let $S := \ker P$, and let $R \subset \operatorname{Sym}^2 L$. If $1_{\operatorname{Sym}^2 L}$ denotes the identity of $\operatorname{Sym}^2 L$, then*

$$\ker((P \otimes 1_{\operatorname{Sym}^2 L})\delta) = \operatorname{Sym}^3 S,$$

and the inverse image of $D \otimes R$ under $(P \otimes 1_{\operatorname{Sym}^2 L})\delta$ has dimension at most

$$\boxed{\binom{\dim L - d + 2}{3} + d \dim R.} \tag{4.1}$$

Proof. The first equality follows in coordinates: all derivatives in directions modulo S vanish precisely when the cubic depends only on S. The inverse image of a subspace has dimension at most the kernel dimension plus the subspace dimension, which gives (4.1). □

Lemma 4.2 (Squarefree projected-symbol table)**.** *Let $\widehat{L} = \mathbb{K}^6$ with basis $z_1, \dots, z_6$, and define the squarefree quadratic and cubic spaces by*

$$\widehat{F} := \operatorname{span}\{z_a z_b : 1 \le a < b \le 6\},$$
$$\widehat{U} := \operatorname{span}\{z_a z_b z_c : 1 \le a < b < c \le 6\}.$$

The restriction of polarization to $\widehat{U}$ is determined by

$$\delta(z_a z_b z_c) = z_a \otimes z_b z_c + z_b \otimes z_a z_c + z_c \otimes z_a z_b.$$

If $P : \widehat{L} \twoheadrightarrow D$ has rank d and $R \subset \widehat{F}$ has dimension at most $r \le 3$, let $q_R : \widehat{F} \twoheadrightarrow \widehat{F}/R$ be the quotient map. Then the following table is a rowwise lower bound for $\operatorname{rank}((P \otimes q_R)\delta)$:

$r \backslash d$	0	1	2	3	4	5	6
0	0	10	16	19	20	20	20
1	0	9	14	16	16	20	20
2	0	8	12	13	16	19	20
3	0	7	10	10	15	17	19.

(4.2)

Proof. The diagonal torus preserves δ. Choose a one-parameter subgroup whose weights on the six variables and the fifteen pair monomials are distinct. The projective limit of $(\ker P, R)$ in the product of Grassmannians consists of coordinate subspaces. The displayed symbol is a map of the tautological

quotient bundles on that product, so its rank can only drop in specialization. Thus $\ker P$ may be taken as the coordinate span indexed by a vertex set $B \subset [6]$, and R as the span of monomials indexed by an edge set $A \subset \binom{[6]}{2}$ with at most r edges. Put $O := [6] \setminus B$. Let $e_A(B)$ be the number of edges of A with both endpoints in B, let $\deg_A(b, O)$ be the number of neighbors of b in O, and let $A[O]$ be the graph induced by A on O.

Distinct source cubics have disjoint output supports. The number of killed cubics is

$$\binom{|B|}{3} + |O|e_A(B) + \sum_{b\in B} \binom{\deg_A(b,O)}{2} + \#\{\text{triangles of } A[O]\}. \tag{4.3}$$

Inspection of edge sets of size at most three gives (4.2). The exact replay independently checks all 45,696 triples (r, B, A), with $0 \le r \le 3$ and $|A| \le r$, and records a minimizer for each entry. □

5. The half-defect estimate

Let $T = \ell_1 \cdots \ell_6 \neq 0$, and set

$$L = \operatorname{span}\{\ell_1, \dots, \ell_6\}, \quad U = \mathcal{D}_3(T), \quad u = \dim U,$$

$$F = \mathcal{D}_2(T), \quad R = F \cap E_2.$$

All derivatives in U are spanned by the twenty products of three of the six factors, so $u \le 20$, also for dependent or repeated factors. For a surjection $P : L \twoheadrightarrow D$ onto a vector space D of dimension d, let $q_R : F \twoheadrightarrow F/R$ be the quotient map and define

$$\beta_{P,R} : U \xrightarrow{\delta} L \otimes F \xrightarrow{P\otimes q_R} D \otimes (F/R).$$

Proposition 5.1 (Half-defect quotient-symbol estimate)**.** *For every nonzero sextic Chow term and every quotient of its actual factor span,*

$$\boxed{\operatorname{rank}\beta_{P,R} + \frac{20-u}{2} \ge \frac{10}{3}d.} \tag{5.1}$$

Proof. Write $\ell = \dim L$. We call $\beta_{P,R}$ a *full symbol* when $d = \ell$, equivalently when P is an isomorphism. Whenever a parenthesized row is displayed below, its entries are indexed from left to right by $d = 0, 1, \dots, \ell$.

Factor span at most four. Choose ℓ independent factors as coordinates. A generic diagonal one-parameter subgroup degenerates the remaining factors to unique coordinate initials, so T specializes to

$$x_1^{m_1} \cdots x_\ell^{m_\ell}, \qquad m_j \ge 1, \qquad \sum_j m_j = 6.$$

Middle rank cannot increase in the limit. Counting degree-three divisors over the positive partitions of six gives

$$\ell = 1, 2, 3, 4 \quad \Longrightarrow \quad u \ge 1, 2, 4, 8. \tag{5.2}$$

Lemma 3.1 gives $\dim R \le 1$. For $0 < d < \ell$, Lemma 4.1 gives

$$\operatorname{rank} \beta_{P,R} \ge \max\left\{0, u - \binom{\ell - d + 2}{3} - d\right\}. \tag{5.3}$$

At $d = \ell$, a kernel cubic lies in $E_2^{(1)} = E_3$, but it is supported on at most six essential variables, contrary to Lemma 2.3. Thus the full symbol is injective. Minimizing over the conservative interval from (5.2) to $\min\{20, \binom{\ell+2}{3}\}$ gives

ℓ	$\operatorname{rank} \beta_{P,R} + (20-u)/2$ for $d = 0, \dots, \ell$
4	$(0, 9/2, 8, 10, 14)$
3	$(5, 15/2, 9, 12)$
2	$(8, 9, 11)$
1	$(19/2, 21/2)$.

(5.4)

Six independent factors. Here $u = 20$, the actual derivative spaces are the squarefree spaces, and $\dim R \le 3$. The last row of (4.2), improved at $d = 6$ by full-symbol injectivity, gives

$$(0, 7, 10, 10, 15, 17, 20). \tag{5.5}$$

Five-dimensional factor span. The unique relation among the six factors puts T, after changing and rescaling coordinates, into exactly one form

$$T_s = x_1x_2x_3x_4x_5(x_1 + \cdots + x_s), \qquad 1 \le s \le 5. \tag{5.6}$$

Lemma 3.1 gives $\dim R \le 1$. Exact differentiation of the actual polynomial gives

s	1	2	3	4	5
$\dim F$	11	11	13	14	15
u	14	14	18	20	20.

(5.7)

The replay constructs the order-four and order-three derivative matrices of (5.6) in the ordinary monomial bases and row-reduces them over $\mathbb{Q}$.

For $s = 3$, the actual space U contains the nine squarefree cubics other than $x_1x_2x_3$. For $s = 4, 5$, it contains all ten squarefree cubics. For a nonzero direction $\sum a_j\partial_{x_j}$, choose $a_j \ne 0$. Differentiating the displayed squarefree cubics containing x_j, and projecting modulo x_jL, gives distinct pair monomials with coefficient a_j. Their number is at least five for $s = 3$ and six for $s = 4, 5$. Thus every nonzero directional map has at least this rank. Quotienting by R loses at most one. Together with (4.1) and full-symbol injectivity this gives

s	u	$\operatorname{rank} \beta_{P,R} + (20-u)/2$ for $d = 0, \dots, 5$
4, 5	20	$(0, 5, 8, 13, 15, 20)$
3	18	$(1, 5, 7, 12, 14, 19)$.

(5.8)

For $s = 1$, the actual U is spanned by the ten squarefree cubics and $x_1^2x_j$, $2 \le j \le 5$. If a direction $\lambda = \sum a_j\partial_{x_j}$ has $a_j \ne 0$ for some $j \ge 2$, the six squarefree cubics divisible by x_j and $x_1^2x_j$, projected modulo x_jL, give seven independent quadrics. The pure x_1 direction has rank ten.

For $s = 2$, a basis consists of the seven squarefree cubics containing at most one of x_1, x_2, the six cubics

$$x_2^2 x_j + 2x_1 x_2 x_j, \qquad x_1 x_2 x_j + \tfrac{1}{2} x_1^2 x_j \quad (3 \le j \le 5),$$

and $x_1 x_2 (x_1 + x_2)$. If $a_j \neq 0$ for some $j \geq 3$, five squarefree inputs and the two special inputs give, modulo $x_j L$, five distinct pair monomials and the independent quadrics $x_2^2 + 2x_1 x_2$ and $x_1 x_2 + \frac{1}{2} x_1^2$. If $\lambda = a_1 \partial_{x_1} + a_2 \partial_{x_2}$, the image contains the three pair monomials on x_3, x_4, x_5, a nonzero direction in each $\operatorname{span}\{x_1 x_j, x_2 x_j\}$, and the nonzero quadratic

$$a_2 x_1^2 + 2(a_1 + a_2) x_1 x_2 + a_1 x_2^2.$$

Thus every nonzero directional map has rank at least seven, and every positive-rank quotient symbol has rank at least six after reducing by R.

For $d = 4$, let $S = \ker P$ be a line and let v lie in the symbol kernel. Its four complementary derivatives span a subspace of the at-most-one-dimensional space R. If one is nonzero, choose a basis of the complementary directions making these derivatives $q, 0, 0, 0$, with $0 \neq q \in R$. In characteristic zero the three vanishing derivatives imply that v depends only on S and one complementary variable. Then q also uses at most two variables, contradicting the rank-four floor of Lemma 2.3. Hence $v \in \operatorname{Sym}^3 S$, so the kernel has dimension at most one. The full symbol at $d = 5$ is injective. With (4.1), the $s = 1, 2$ row is

$$(3, 9, 9, 10, 16, 17). \tag{5.9}$$

Every entry of (5.4), (5.5), (5.8), and (5.9) is at least $10d/3$. This proves (5.1). □

Remark 5.2 (Actual derivative spaces)**.** The formal pair- and triple-product spans for T_2 have dimensions 12 and 17, while the actual derivative spaces have dimensions 11 and 14. For T_3, the formal triple-product span has dimension 19 while the actual middle derivative space has dimension 18. The proof above deliberately uses the formal squarefree table only in the six-independent-factor case. The unequal dimensions are retained as regression tests.

6. A symmetric image-span inequality

Lemma 6.1. *Let $A_1, \dots, A_N : W^* \to W$ be linear maps that are symmetric under the natural pairing between W^* and W. Set $r_i := \operatorname{rank} A_i$ and $D := \dim(\sum_{i=1}^N \operatorname{im} A_i)$. Then*

$$\operatorname{rank}\Big(\sum_i A_i\Big) \geq 2D - \sum_i r_i. \tag{6.1}$$

Proof. Choose an injective map $B_i : \mathbb{K}^{r_i} \to W$ with image $\operatorname{im} A_i$, and let $B_i^* : W^* \to (\mathbb{K}^{r_i})^*$ be its dual map. Symmetry gives $\ker A_i = (\operatorname{im} A_i)^\perp$, so $A_i = B_i J_i B_i^*$ for an invertible symmetric map $J_i : (\mathbb{K}^{r_i})^* \to \mathbb{K}^{r_i}$. Let

$B : \bigoplus_i \mathbb{K}^{r_i} \to W$ be the map with blocks B_i, and let $J := \bigoplus_i J_i$. Sylvester's inequality gives

$$\operatorname{rank}(BJB^*) \geq \operatorname{rank}(BJ) + \operatorname{rank}(B^*) - \sum_i r_i = 2D - \sum_i r_i.$$

□

7. The global lower bound

Suppose that N is a positive integer and

$$\operatorname{perm}_6 = \sum_{i=1}^{N} T_i$$

with no zero summands. Let

$$A_i = C_{3,3}(T_i), \quad U_i = \operatorname{im} A_i, \quad u_i = \dim U_i, \quad \delta_i = 20 - u_i, \quad \Delta = \sum_i \delta_i.$$

Put $U = \sum_i U_i$ and $\dim U = 400 + h$. The sum of the symmetric maps A_i is the permanent middle catalecticant, with rank 400 and image E_3. Lemma 6.1 gives

$$\boxed{h \leq 10N - 200 - \frac{\Delta}{2}.} \tag{7.1}$$

Let $F_i := \mathcal{D}_2(T_i)$, $F := \sum_i F_i$, and $Q := F/E_2$, and let $q : F \twoheadrightarrow Q$ be the quotient map. Linearity of the derivative spaces gives $E_2 \subset F$. If $L_i := \operatorname{span}\{\ell_{i1}, \ldots, \ell_{i6}\}$ is the actual factor span of $T_i = \ell_{i1} \cdots \ell_{i6}$, define

$$\widetilde{\beta} : \bigoplus_{i=1}^{N} U_i \longrightarrow V \otimes Q, \qquad (g_i)_{i=1}^{N} \longmapsto \sum_{i=1}^{N} (\iota_i \otimes q|_{F_i}) \delta_{L_i}(g_i),$$

where $\iota_i : L_i \hookrightarrow V$ is inclusion. Its kernel consists precisely of tuples for which every first derivative of $\sum_i g_i$ lies in E_2, equivalently $\sum_i g_i \in E_3$ by Lemma 2.3. Since $E_3 \subset U$,

$$\operatorname{rank} \widetilde{\beta} = \dim(U/E_3) = h. \tag{7.2}$$

Since $E_2 \subset F$, $\partial E_2 = V$, and $\partial F_i \subset L_i$, the spaces L_i span all 36 variables. Choose any ordering and put

$$W_0 := 0, \qquad W_i := \sum_{j=1}^{i} L_j, \qquad d_i = \dim(W_i/W_{i-1}), \qquad \sum_i d_i = 36.$$

Let $Z_i := \operatorname{im}((1_{L_i} \otimes q|_{F_i})\delta_{L_i}) \subset L_i \otimes Q$, where 1_{L_i} is the identity map of L_i. Set $H_0 := 0$ and $H_i := \sum_{j=1}^{i} Z_j$. The quotient projection

$$\pi_i : W_i \otimes Q \longrightarrow (W_i/W_{i-1}) \otimes Q$$

kills H_{i-1}, hence

$$\dim H_i - \dim H_{i-1} \geq \dim \pi_i(H_i) = \dim \pi_i(Z_i). \tag{7.3}$$

On the current block, the first-factor map is

$$P_i : L_i \twoheadrightarrow L_i/(L_i \cap W_{i-1}), \qquad \operatorname{rank} P_i = d_i.$$

The natural map

$$F_i/(F_i \cap E_2) \longrightarrow F/E_2$$

is injective. Therefore $\dim \pi_i(Z_i)$ is exactly the local symbol rank in Proposition 5.1. Summing (7.3) gives

$$\boxed{h \geq \sum_i \left(\frac{10}{3}d_i - \frac{\delta_i}{2}\right) = 120 - \frac{\Delta}{2}.} \tag{7.4}$$

Comparing (7.1) and (7.4) cancels the complete individual middle-rank defect and yields $N \geq 32$.

For the reverse inequality, Glynn's identity [2] gives

$$\operatorname{perm}_6 = \frac{1}{32} \sum_{\epsilon_1=1,\ \epsilon_2,\ldots,\epsilon_6=\pm 1} \left(\prod_{r=1}^{6} \epsilon_r\right) \prod_{c=1}^{6} \left(\sum_{r=1}^{6} \epsilon_r x_{rc}\right).$$

Walsh cancellation leaves exactly the monomials that use each row once. This is a 32-term Chow decomposition. Theorem 2.2 follows. For a characteristic-zero field that is not algebraically closed, any hypothetical shorter decomposition remains one over its algebraic closure, where the lower bound just proved applies. The displayed upper bound is defined over $\mathbb{Q}$. Thus the equality holds over that field as well.

8. Computational lemmas and certificate contract

A *certificate contract* in this section consists of a finite input class, a specified exact arithmetic domain, an asserted output, and a verifier that exits unsuccessfully unless it reconstructs that output. Vertices of the complete graph K_6 and the two color classes of the bipartite graph $K_{6,6}$ are both ordered by [6]. Subsets and monomials are ordered lexicographically. All ranks are exact ranks over $\mathbb{Q}$.

Lemma 8.1 (Coordinate intersection certificate)**.** *Among coordinate subspaces spanned by e variables of V, for $0 \leq e \leq 6$, the maximum number of 2-by-2 permanent quadrics supported on the subspace is*

$$(0, 0, 0, 0, 1, 1, 3).$$

Certificate proof. A coordinate subspace is identified with an e-edge subgraph of $K_{6,6}$, and a supported rectangle permanent is identified with a four-cycle. The primary replay enumerates row-permutation representatives. The independent program `anc/original/computation/scripts/n6_independent_finite_core_verification.py` instead enumerates every labelled edge subset, namely $\sum_{e=0}^{6} \binom{36}{e} = 2{,}391{,}496$ graphs, and directly counts pairs of common neighbors. Both computations give the displayed maxima. □

Lemma 8.2 (Squarefree-symbol certificate)**.** *Let* $0 \le r \le 3$ *and* $0 \le d \le 6$ *have the meanings fixed in Section 4: r bounds the number of killed edge coordinates and d is the quotient dimension. The minimum coordinate squarefree-symbol ranks are*

$r\backslash d$	0	1	2	3	4	5	6
0	0	10	16	19	20	20	20
1	0	9	14	16	16	20	20
2	0	8	12	13	16	19	20
3	0	7	10	10	15	17	19

and the exhaustive coordinate domain contains 45,696 *cases.*

Certificate proof. For each coordinate kernel and each killed-edge set of cardinality at most r, the verifier visits all twenty three-subsets of [6]. A cubic coordinate survives exactly when one of its three vertex-edge factorizations survives. The primary replay also evaluates the closed killed-column formula; the independent computation uses only this direct survival test and does not import the primary module or read the frozen JSON. It reproduces the table and the case count exactly. □

Lemma 8.3 (Normal-form and half-defect certificate)**.** *For the five dependent-factor normal forms in Section 5, the exact derivative-dimension pairs are*

$$(11, 14), (11, 14), (13, 18), (14, 20), (15, 20).$$

The exact low-factor-span floors are $1, 2, 4, 8$*, and all half-defect rows listed in Section 5 satisfy the required inequality. Consequently the finite local input to the cancellation in Section 7 is valid.*

Certificate proof. The primary verifier expands each normal form as a sparse polynomial with rational coefficients, differentiates it symbolically, and row-reduces the actual derivative coefficient matrices. It separately enumerates the positive partitions of six into at most four parts to obtain the monomial floors, then evaluates every displayed half-defect row as a rational number. In particular, it tests actual derivative spaces rather than substituting formal pair- or triple-product spans. A missing profile, an unequal dimension, or a failed inequality terminates the replay with an error. This lemma is independently specified here and is also checked by the standard-library program `anc/original/computation/scripts/n6_dependent_normal_forms_independent.py`, which does not import the primary verifier or read its frozen JSON. □

These three lemmas state the mathematical content supplied by computation. The fixed-point reductions, prolongation identity, directional cases, symmetric image-span inequality, filtration increment, and defect cancellation remain written arguments in the preceding sections.

9. Exact verification

From the extracted submission root, run

```
python -B anc/verify_submission.py
```

This requires Python 3.11 or newer and only its standard library. It checks the exact computational subset against the unchanged September manifest and executes the original complete `anc/original/verify_all.py` entry point. The full historical package remains in the public repository; the ancillary README lists the omitted historical manuscript files, which are not computation inputs. The original primary verifier derives the coordinate intersection maxima, all 45,696 coordinate cases behind (4.2), the actual normal-form derivative ranks and squarefree subspaces in (5.7)–(5.8), the low-span monomial floors, and the final cancellation. It also verifies the unequal formal and actual dimensions in Remark 5.2.

The nine unit tests exercise the primary implementation as regression tests. The finite-core program is independent of that implementation and of the reference JSON; its scope is exactly Lemmas 8.1 and 8.2. The normal-form program independently reconstructs the normal-form derivative ranks used in Lemma 8.3. A further independent integer expansion of Glynn's identity checks all $32 \cdot 6^6 = 1{,}492{,}992$ row choices, with exactly 720 surviving permutation monomials. The wrapper writes fresh outputs only to temporary directories; it does not overwrite frozen evidence. For a repository checkout the original public reproduction command remains `python -B perm6/verify_all.py`.

The torus-specialization reductions, prolongation identity, directional arguments, symmetric image-span lemma, and filtration increment (7.3) are written proofs rather than numerical inferences.

Remark 9.1 (Scope)**.** The theorem concerns ordinary Chow rank over characteristic zero. It does not determine border Chow rank or prove the analogous equality $\mathrm{R}_{\mathrm{Chow},\mathbb{K}}(\mathrm{perm}_n) = 2^{n-1}$ for the permanent perm_n of a generic n by n matrix. Product rank and Chow rank are used synonymously in the relevant literature; see [3].

Disclosure of AI-assisted tools

ChatGPT was used to help audit and develop arguments, revise the exposition, and prepare verification and submission materials. Responsibility for the final manuscript rests with the author.

References

[1] A. Borel, *Linear algebraic groups*, second ed., Graduate Texts in Mathematics, vol. 126, Springer-Verlag, New York, 1991, doi:10.1007/978-1-4612-0941-6.

[2] D. G. Glynn, *The permanent of a square matrix*, European J. Combin. **31** (2010), no. 7, 1887–1891, doi:10.1016/j.ejc.2010.01.010.

[3] N. Ilten and Z. Teitler, *Product ranks of the* 3×3 *determinant and permanent*, Canad. Math. Bull. **59** (2016), no. 2, 311–319, doi:10.4153/CMB-2015-076-1.